

CSE-606-SOFTWARE-ENGINEERING

Connecting patients with the right therapists for personalized mental health care.

Getting Started

1. Clone the repository and change your working directory to the project root.
2. Make sure you have installed Python on your system.
3. Create a virtual environment:

```
python3 -m venv venv'
```

4. Activate it:

```
source venv/bin/activate
```

5. Install dependencies:

```
pip install -r requirements.txt
```

6. Run the web server:

```
python main.py
```

7. Open `localhost:8000` in your browser!

First Iteration: Prototype

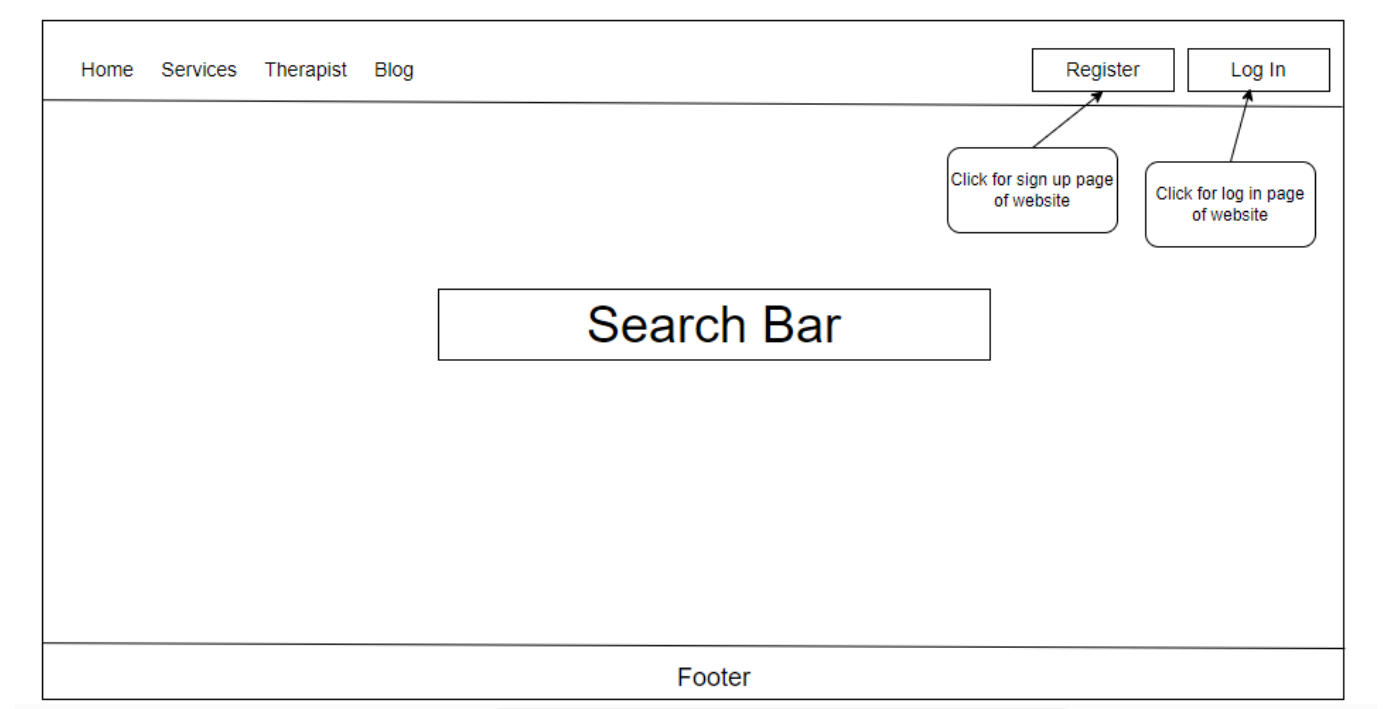
1. Requirements: Description with UI sketch of main use cases.

Main Use Cases

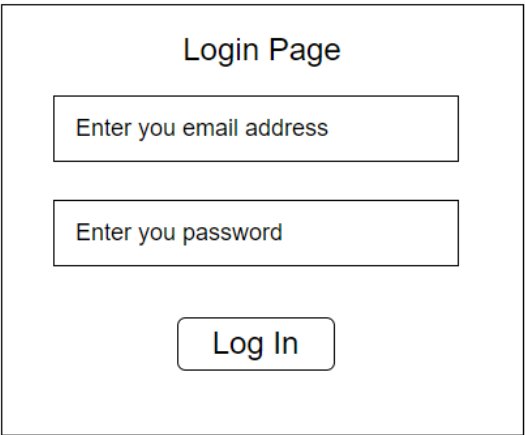
1. A user (patient or therapist) should be able to signup.
2. A user (patient or therapist) should be able to login & logout, i.e. session auth using web cookies.
3. A patient should be able to search among therapists.
4. Users should be able to complete/view/edit their profile.

Sketches

Home Page:



Login Page:



Signup Page:

Signup Page

Enter your Full Name

Enter your Email Address

Enter your Password

Re-enter your Password

☐

 Patient

☐

 Therapist

Sign Up

Therapists:

Home Services Therapist Blog

Register

Log In

Therapists

Therapist 1

Therapist 2

Therapist 3

Therapist 4

Footer

Patient Profile:

HomeServicesTherapistBlog

Log Out

Patient's Profile

Name:

Date of Birth:

Gender:

Contact Information:

Emergency Contact Information:

Medical History:

Insurance Information:

Save/Edit Profile

Footer

Therapist Profile:

HomeServicesTherapistBlog

Log Out

Therapist's Profile

Name:

Date of Birth:

Gender:

Contact Information:

License/Certification:

Specialization:

Experience(in years):

Save/Edit Profile

Footer

2. Database design: Description of data entities and relationships, entity-relationship diagram, SQL code to design database, sample date.

Data Entities and Relationships:

User

- Attributes: id (primary key), name, email, password, is_patient, created_at
- Relationships: One-to-One with Profile (user <-> profile)

Profile (Abstract Class)

- Attributes: id (primary key), user (foreign key), dob, gender, contact_info
- Relationships: One-to-One with User (profile <-> user)

PatientProfile

- Attributes: id (primary key), emergency_contact_info, medical_history, insurance_info
- Inheritance: Inherits from Profile

TherapistProfile

- Attributes: id (primary key), license_info, speciality, experience, bio
- Inheritance: Inherits from Profile

ERD

[frontend/sketches/ER.png](#)

SQL Code to Create Tables

```
CREATE TABLE "user" (  
  "id" INTEGER NOT NULL PRIMARY KEY,  
  "name" VARCHAR,  
  "email" VARCHAR UNIQUE,  
  "password" VARCHAR,  
  "is_patient" BOOLEAN DEFAULT 1,  
  "created_at" DATETIME DEFAULT (DATETIME('now'))  
);  
  
CREATE TABLE "patientprofile" (  
  "id" INTEGER NOT NULL PRIMARY KEY,  
  "user_id" INTEGER UNIQUE,  
  "dob" DATE,  
  "gender" VARCHAR,  
  "contact_info" VARCHAR,  
  "emergency_contact_info" VARCHAR,  
  "medical_history" VARCHAR,  
  "insurance_info" VARCHAR,  
  FOREIGN KEY ("user_id") REFERENCES "user" ("id")  
);  
  
CREATE TABLE "therapistprofile" (  
  "id" INTEGER NOT NULL PRIMARY KEY,  
  "user_id" INTEGER UNIQUE,  
  "dob" DATE,  
  "gender" VARCHAR,  
  "contact_info" VARCHAR,  
  "license_info" VARCHAR,  
  "speciality" VARCHAR,  
  "experience" DECIMAL(3, 1),  
  "bio" VARCHAR,
```

```
FOREIGN KEY ("user_id") REFERENCES "user" ("id")
);
```

SQL Code to Add Sample Data

```
-- Sample User Data
INSERT INTO "user" ("name", "email", "password", "is_patient",
"created_at")
VALUES
    ('John Doe', 'john@example.com', 'password123', 1, '2023-10-01
08:00:00'),
    ('Dr. Smith', 'dr.smith@example.com', 'securepass', 0, '2023-10-01
09:15:00');

-- Sample PatientProfile Data
INSERT INTO "patientprofile" ("user_id", "dob", "gender", "contact_info",
"emergency_contact_info", "medical_history", "insurance_info")
VALUES
    (1, '1990-05-15', 'Male', '123-456-7890', 'Jane Doe - 987-654-3210',
'Allergies: Pollen', 'ABC Insurance Co.');
```

```
-- Sample TherapistProfile Data
INSERT INTO "therapistprofile" ("user_id", "dob", "gender",
"contact_info", "license_info", "speciality", "experience", "bio")
VALUES
    (2, '1975-03-20', 'Female', '987-654-3210', 'License #12345',
'Psychologist', 10.5, 'I specialize in cognitive-behavioral therapy.');
```

3. Architectural design: Description of 3-tier architecture (e.g., UI, Logic, DB).

In this project, the 3-tier architecture can be described as follows:

Presentation Tier: This tier is responsible for handling user interactions and displaying information to the user. In this project, the presentation tier is implemented using HTML, CSS, and JavaScript to create the user interface.

Application Tier: This tier is responsible for processing user requests and generating responses. In this project, the application tier is implemented using Pure Python to handle HTTP requests and responses.

Data Tier: This tier is responsible for storing and retrieving data. In this project, the data tier is implemented using a SQLite database and PeeWee library to store user and therapist information.

The presentation tier communicates with the application tier using HTTP requests and responses. The application tier communicates with the data tier using SQL queries to retrieve and store data in the database.

This 3-tier architecture provides a separation of concerns between the different layers of the application, making it easier to maintain and scale the application.

4. A runnable prototyping with GUI, database (if needed).

All codes are in this repository. Read the **Getting Started** section in order to run the web app. Also, you can watch the video in the following section.

5. Video recordings of user acceptance tests.

YouTube Link: <https://youtu.be/tLZCMh9D0RU>

Project File Structure

```
.
├── LICENSE
├── README.md
├── __init__.py
├── controllers
│   ├── __init__.py
│   └── users.py
├── database.db
├── frontend
│   ├── README.md
│   ├── home.html
│   ├── login.html
│   ├── patient_profile.html
│   ├── patient_profile_edit.html
│   ├── sketches
│   │   ├── source_sketches.drawio
│   │   ├── home.PNG
│   │   ├── login.PNG
│   │   ├── profile_patient.PNG
│   │   ├── profile_therapist.PNG
│   │   ├── signup.PNG
│   │   └── therapists.PNG
│   ├── therapist_profile.html
│   ├── therapist_profile_edit.html
│   └── therapists.html
├── main.py
├── mini_web.py
├── models
│   ├── __init__.py
│   └── users.py
├── requirements.txt
├── static
│   ├── css
│   │   └── bootstrap.css
│   ├── images
│   └── js
│       ├── bootstrap.js
│       └── script.js
├── templates
│   ├── about.html
│   └── base.html
```

```
11 directories, 45 files
```

- ☐ Adding profile photos.
- ☐ Therapist/Patient detail view.
- ☐ Adding scheduling system.
- ☐ Adding review (feedback) system.
- ☐ Better search algorithm using filters.
- ☐ AI recommendation system.